# Linear Algebra and Its Applications

## Section 7.3: Constrained Optimization

Detailed worked solutions

Mathjock

**Text and numbering.** David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

**Source scope.** The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

**How to read the solutions.** Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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## 7.3 Constrained optimization

A quadratic form is an expression $Q(x) = x^T Ax$, where we take $A$ to be symmetric. A coefficient of $x_i x_j$ with $i \neq j$ must be divided equally between $a_{ij}$ and $a_{ji}$. For example, $6x_1x_2$ means $a_{12} = a_{21} = 3$, not 6.

**The central argument.** The spectral theorem gives an orthonormal eigenbasis $u_1, \ldots, u_n$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Put $P = [u_1 \ \cdots \ u_n]$ and $x = Py$. Then $P^TP = I$, so

$$x^Tx = y^Ty, \qquad Q(x) = \sum_{j=1}^{n} \lambda_j y_j^2.$$

On the unit sphere, the nonnegative numbers $y_j^2$ add to one. Thus

$$\lambda_n \leq Q(x) \leq \lambda_1.$$

Taking $x = u_1$ attains the upper bound. If we also require $x^Tu_1 = 0$, then $y_1 = 0$, and the maximum becomes $\lambda_2$, attained at $u_2$. Eigenvalues here are listed *with multiplicities*; a repeated largest eigenvalue can remain the largest value after one direction is excluded. These observations explain the optimization steps below without requiring multivariable differentiation.

### Practice problem 7.3.P1: Remove the mixed term using an orthonormal basis

**What we must show.** Diagonalize $3x_1^2 + 3x_2^2 + 2x_1x_2$ by $x = Py$. The symmetric matrix is $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Its characteristic polynomial is $(3-\lambda)^2 - 1$, with roots $4, 2$. For 4, solving $(A - 4I)v = 0$ gives $v_2 = v_1$; choose $(1,1)^T$. For 2, the equations give $v_2 = -v_1$; choose $(-1,1)^T$. Each vector has length $\sqrt{2}$, so use

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Its columns have length one and dot product zero. Therefore $P^TAP = \operatorname{diag}(4, 2)$, and $Q(Py) = 4y_1^2 + 2y_2^2$.

**Conclusion.** The displayed $P$ gives the requested substitution. Normalizing the eigenvectors is essential to making $P$ orthogonal.

### Practice problem 7.3.P2: Translate the maximizing direction back to the original variables

For a unit $x$, the preceding change of variables gives $y_1^2 + y_2^2 = 1$. Consequently $Q(x) = 4y_1^2 + 2y_2^2 \leq 4$. Equality holds for $y = (1,0)^T$. But the requested vector is $x$, not $y$: multiplying by $P$ gives $x = (1,1)^T/\sqrt{2}$. Its squared length is $(1+1)/2 = 1$.

**Conclusion.** The maximum is 4, attained at $(1,1)^T/\sqrt{2}$. Its negative also attains the same value.

### Exercise 7.3.1: An orthogonal substitution in three variables

**What we must show.** Obtain the diagonal coefficients $9, 6, 3$ for the given quadratic form.

**Build the matrix and find its eigenvectors.**

The coefficients give

$$A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & -2 \\ 0 & -2 & 7 \end{bmatrix}.$$

Expanding $\det(\lambda I - A)$ gives $(\lambda - 9)(\lambda - 6)(\lambda - 3)$. Solving the corresponding homogeneous systems gives the convenient eigenvectors

$$v_9 = (-1, -2, 2)^T, \quad v_6 = (2, 1, 2)^T, \quad v_3 = (-2, 2, 1)^T.$$

For example, at $\lambda = 9$ the first equation is $-4a + 2b = 0$, so $b = 2a$; the last is $-2b - 2c = 0$, so $c = -b$. Taking $a = -1$ produces $v_9$. The other two vectors are checked by multiplying by $A$. Each has squared length 9, and their pairwise dot products are zero.

**Normalize and put the columns in the requested order.**

Set

$$P = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Then $P^T P = I$ and $AP = P\,\mathrm{diag}(9, 6, 3)$. Multiplying this last identity on the left by $P^T$ gives $P^T AP = \mathrm{diag}(9, 6, 3)$.

**Conclusion.** With $x = Py$, the form is $9y_1^2 + 6y_2^2 + 3y_3^2$.

## Exercise 7.3.2: Include the missing zero coefficient

**What we must show.** Obtain the stated diagonal form, keeping three coordinates. The symmetric matrix is

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}, \qquad \det(\lambda I - A) = \lambda(\lambda - 2)(\lambda - 5).$$

Corresponding eigenvectors are $(1, 1, 1)^T$, $(-2, 1, 1)^T$, and $(0, -1, 1)^T$, for $5, 2, 0$, respectively. Their squared lengths are $3, 6, 2$, and their pairwise dot products vanish. Thus take

$$P = \begin{bmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}.$$

Direct multiplication gives $P^T AP = \mathrm{diag}(5, 2, 0)$.

**Conclusion.** $x = Py$ transforms the form to $5y_1^2 + 2y_2^2 + 0y_3^2$. The zero coefficient does not remove a coordinate from the substitution.

## Exercise 7.3.3: Exclude one principal direction

**(a) Bound the form on all unit vectors.**

Exercise 7.3.1 gives $Q(x) = 9y_1^2 + 6y_2^2 + 3y_3^2$ and $y_1^2 + y_2^2 + y_3^2 = 1$. Replacing 6 and 3 by 9 shows $Q(x) \le 9$.

**(b) Give a maximizing unit vector in the original coordinates.**

Take $u = (-1, -2, 2)^T/3$, the first column of $P$. Then $Au = 9u$ and $u^T u = 1$, so $Q(u) = u^T(9u) = 9$.

**(c) Impose the orthogonality condition.**

Because $x = Py$ and $u = Pe_1$, we have $x^T u = y^T e_1 = y_1$. Thus $x^T u = 0$ forces $y_1 = 0$. Now $Q(x) = 6y_2^2 + 3y_3^2 \le 6(y_2^2 + y_3^2) = 6$. Equality holds at the second column $(2, 1, 2)^T/3$ of $P$.

**Conclusion.** The answers are 9, $u = (-1, -2, 2)^T/3$, and 6.

### Exercise 7.3.4: The same constraints with a null direction

Use the orthogonal coordinates of Exercise 7.3.2. On the unit sphere, $Q(x) = 5y_1^2 + 2y_2^2 \le 5(y_1^2 + y_2^2 + y_3^2) = 5$. The unit vector $u = (1, 1, 1)^T/\sqrt{3}$ has $Au = 5u$, so it attains 5. The additional equation $x^T u = 0$ is $y_1 = 0$. Therefore $Q(x) = 2y_2^2 \le 2$, with equality at $(-2, 1, 1)^T/\sqrt{6}$.

**Conclusion.** (a) 5; (b) $u = (1, 1, 1)^T/\sqrt{3}$; (c) 2.

### Exercise 7.3.5: A negative mixed coefficient

The matrix is $A = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$. The equation $(5 - \lambda)^2 - 4 = 0$ gives eigenvalues $7, 3$. Their unit eigenvectors are $u = (-1, 1)^T/\sqrt{2}$ and $v = (1, 1)^T/\sqrt{2}$. They are perpendicular. Writing $x = au + bv$ gives $\|x\|^2 = a^2 + b^2$ and $Q(x) = 7a^2 + 3b^2$. Hence the unrestricted unit maximum is 7, attained at $u$. When $x^T u = 0$, we have $a = 0$ and $b^2 = 1$, so the value is 3.

**Conclusion.** (a) 7; (b) $(-1, 1)^T/\sqrt{2}$; (c) 3.

### Exercise 7.3.6: Remember to halve an odd mixed coefficient

Here $A = \begin{bmatrix} 7 & 3/2 \\ 3/2 & 3 \end{bmatrix}$, since the mixed term is $3x_1x_2$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - \frac{9}{4} = 0,$$

whose roots are $15/2$ and $5/2$. At $15/2$, the first row of $(A - \lambda I)v = 0$ gives $-v_1 + 3v_2 = 0$; choose $(3, 1)^T$. Its unit version is $u = (3, 1)^T/\sqrt{10}$. A perpendicular unit eigenvector is $v = (-1, 3)^T/\sqrt{10}$, for $5/2$. For $x = au + bv$, the unit constraint gives $a^2 + b^2 = 1$, and $Q(x) = (15/2)a^2 + (5/2)b^2$. Requiring $x^T u = 0$ sets $a = 0$.

**Conclusion.** (a) $15/2$; (b) $(3, 1)^T/\sqrt{10}$; (c) $5/2$.

### Exercise 7.3.7: Normalize the largest-eigenvalue vector

The given coefficients give

$$A = \begin{bmatrix} -2 & 2 & 0 \\ 2 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix}.$$

Its eigenvalues are $2, -1, -4$, so the maximum on the unit sphere is 2. To find a maximizing vector, solve $(A - 2I)v = 0$. The first row gives $-4v_1 + 2v_2 = 0$, hence $v_2 = 2v_1$. The last gives $2v_2 - 2v_3 = 0$, hence $v_3 = v_2$. Taking $v_1 = 1$ gives $v = (1, 2, 2)^T$, with length 3. Consequently $x = v/3$ is unit, and $Q(x) = x^T(2x) = 2$.

**Conclusion.** One required vector is $(1, 2, 2)^T/3$.

### Exercise 7.3.8: A repeated largest eigenvalue

The matrix is

$$A = \begin{bmatrix} 7 & -4 & -2 \\ -4 & 1 & -4 \\ -2 & -4 & 7 \end{bmatrix}.$$

Its eigenvalues are $9, 9, -3$. At 9, every row equation reduces to $v_1 + 2v_2 + v_3 = 0$. Choose $v = (-1, 0, 1)^T$ in this plane and normalize: $x = (-1, 0, 1)^T/\sqrt{2}$. The squared length is one, $Ax = 9x$, and $Q(x) = 9$. No unit vector can exceed the largest eigenvalue.

**Conclusion.** $(-1, 0, 1)^T/\sqrt{2}$ is a maximizing unit vector; the maximum is 9.

### Exercise 7.3.9: Find the maximum without finding a vector

The mixed coefficient $-2$ gives $A = \begin{bmatrix} 7 & -1 \\ -1 & 3 \end{bmatrix}$. Its characteristic equation is

$$(7 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 10\lambda + 20 = 0.$$

The quadratic formula gives $\lambda = 5 \pm \sqrt{5}$. By the unit-sphere bound established at the start, the maximum is the larger root. No eigenvector is needed for the question asked.

**Conclusion.** The maximum is $5 + \sqrt{5}$.

### Exercise 7.3.10: A form with both positive and negative values

Now $A = \begin{bmatrix} -3 & -1 \\ -1 & 5 \end{bmatrix}$. The equation is

$$(-3 - \lambda)(5 - \lambda) - 1 = \lambda^2 - 2\lambda - 16 = 0,$$

with roots $1 \pm \sqrt{17}$. The constraint is exactly the unit-sphere constraint, so the larger eigenvalue gives the maximum.

**Conclusion.** The maximum is $1 + \sqrt{17}$.

### Exercise 7.3.11: Evaluate a form at a unit eigenvector

If $Ax = 3x$ and $x^T x = 1$, substitute the eigenvector equation first:

$$x^T Ax = x^T(3x) = 3(x^T x) = 3.$$

This particular calculation does not require symmetry of $A$; the two stated equations are sufficient.

**Conclusion.** The value is 3.

### Exercise 7.3.12: Every eigenvalue lies between the extreme unit values

Let $\lambda$ be an eigenvalue of the real symmetric matrix $A$. The spectral theorem supplies a nonzero real eigenvector $v$. Normalize it: $x = v/\|v\|$. Then $x^T x = 1$ and $Ax = \lambda x$, so $x^T Ax = \lambda x^T x = \lambda$. By definition, $m$ is a lower bound and $M$ is an upper bound for the values of the form at all unit vectors. They therefore bound this particular value.

**Conclusion.** $m \le \lambda \le M$. Normalization places the eigenvector on the very set over which the extrema are defined.

### Exercise 7.3.13: Every intermediate value is attained

**What we must show.** Given $m \le t \le M$, construct a unit $x$ with $x^T Ax = t$.

**First separate the constant case.**

If $M = m$, then $t = m$ and any unit eigenvector for $m$ works. The proposed expression can use $\alpha = 0$, giving $x = u_n$. This also handles the one-dimensional case without demanding two perpendicular vectors in a one-dimensional space.

**For $M > m$, choose the interpolation weight explicitly.**

Put $\alpha = (t - m)/(M - m)$. The inequalities on $t$ imply $0 \le \alpha \le 1$, and rearrangement gives $t = (1 - \alpha)m + \alpha M$. Let $Au_1 = Mu_1$ and $Au_n = mu_n$, where the vectors are unit. Distinct eigenvalues of a symmetric matrix have orthogonal eigenvectors: indeed $u_1^T Au_n = mu_1^T u_n$ and, by symmetry, the same expression equals $(Au_1)^T u_n = Mu_1^T u_n$. Since $M - m \neq 0$, $u_1^T u_n = 0$.

**Check both the constraint and the desired value.**

Define $x = \sqrt{1 - \alpha}\, u_n + \sqrt{\alpha}\, u_1$. Expanding gives

$$x^T x = (1 - \alpha)\underbrace{u_n^T u_n}_{1} + \alpha \underbrace{u_1^T u_1}_{1} + 2\sqrt{\alpha(1 - \alpha)}\underbrace{u_n^T u_1}_{0} = 1.$$

The mixed terms in $x^T Ax$ also vanish, because, for example, $u_n^T Au_1 = Mu_n^T u_1 = 0$. The remaining terms give

$$x^T Ax = (1 - \alpha)m + \alpha M = t.$$

**Conclusion.** Every number in $[m, M]$ occurs. The square roots in $x$ make the squared coefficients, rather than the coefficients, add to one.

### Exercise 7.3.14: An exact computer-assisted four-dimensional example

**Convert the mixed terms to a symmetric matrix.**

$$A = \frac{1}{2}\begin{bmatrix} 0 & 1 & 3 & 30 \\ 1 & 0 & 30 & 3 \\ 3 & 30 & 0 & 1 \\ 30 & 3 & 1 & 0 \end{bmatrix}.$$

To check its spectrum without trusting rounded output, use the four pairwise perpendicular vectors

$$(1,1,1,1)^T, \quad (1,-1,-1,1)^T, \quad (-1,1,-1,1)^T, \quad (-1,-1,1,1)^T.$$

Multiplication by $A$ gives these same vectors times $17, 13, -14, -16$, respectively. Each has length 2; dividing them by 2 gives an orthonormal eigenbasis and verifies the entire spectrum.

**Answer the three optimization questions.**

The maximum on unit vectors is 17, attained at $u = (1,1,1,1)^T/2$. Orthogonality to $u$ removes its coordinate, leaving the next largest eigenvalue, 13. It is attained at $(1,-1,-1,1)^T/2$.

**Conclusion.** (a) 17; (b) $(1,1,1,1)^T/2$; (c) 13.

### Exercise 7.3.15: A negative constrained maximum

$$A = \frac{1}{2}\begin{bmatrix} 0 & 3 & 5 & 7 \\ 3 & 0 & 7 & 5 \\ 5 & 7 & 0 & 3 \\ 7 & 5 & 3 & 0 \end{bmatrix}.$$

The same four orthogonal sign vectors from Exercise 7.3.14 are eigenvectors, now with eigenvalues $15/2, -1/2, -5/2, -9/2$, in that order. For instance, the first row sum is $(3+5+7)/2 = 15/2$, verifying the first coordinate of the all-ones eigenvector equation; the other rows have the same sum. Checking the other three sign vectors verifies the remaining eigenvalues. Their unit versions form a basis, so no eigenvalue is omitted. Thus the unit-sphere maximum is $15/2$, at $u = (1,1,1,1)^T/2$. On $u^\perp$, the largest remaining eigenvalue is $-1/2$, attained at $(1,-1,-1,1)^T/2$. A maximum need not be positive.

**Conclusion.** (a) $15/2$; (b) $(1,1,1,1)^T/2$; (c) $-1/2$.

### Exercise 7.3.16: One diagonal term and six mixed terms

The symmetric matrix and a factored characteristic polynomial are

$$A = \begin{bmatrix} 4 & -3 & -5 & -5 \\ -3 & 0 & -3 & -3 \\ -5 & -3 & 0 & -1 \\ -5 & -3 & -1 & 0 \end{bmatrix},$$

$$\det(\lambda I - A) = (\lambda - 9)(\lambda - 3)(\lambda - 1)(\lambda + 9).$$

For an exact check, vectors $(-2,0,1,1)^T$, $(1,-3,1,1)^T$, $(0,0,-1,1)^T$, and $(1,1,1,1)^T$ have eigenvalues $9, 3, 1, -9$, respectively. Each assertion follows by one matrix–vector multiplication, and the vectors

are nonzero and pairwise perpendicular. The first has length $\sqrt{6}$, so choose $u = (-2, 0, 1, 1)^T/\sqrt{6}$. Then $Q(u) = 9$. With $x \perp u$, the maximum becomes 3, attained at $(1, -3, 1, 1)^T/\sqrt{12}$.

**Conclusion.** (a) 9; (b) $(-2, 0, 1, 1)^T/\sqrt{6}$; (c) 3.

### Exercise 7.3.17: The largest eigenvalue can itself be negative

Here

$$A = \begin{bmatrix} -6 & -2 & -2 & -2 \\ -2 & -10 & 0 & 0 \\ -2 & 0 & -13 & 3 \\ -2 & 0 & 3 & -13 \end{bmatrix}.$$

Its characteristic polynomial is $(\lambda + 4)(\lambda + 10)(\lambda + 12)(\lambda + 16)$. Nonzero eigenvectors for $-4, -10, -12, -16$, respectively, are

$$(-3, 1, 1, 1)^T, \quad (0, -2, 1, 1)^T, \quad (1, 1, 1, 1)^T, \quad (0, 0, -1, 1)^T.$$

They are pairwise perpendicular. Checking $Av = \lambda v$ for each one verifies the spectrum independently of an eigenvalue routine. The largest eigenvalue is $-4$, since $-4 > -10 > -12 > -16$; one does *not* choose the largest absolute value. The first vector has squared length 12. After excluding its unit direction, $-10$ is the largest remaining eigenvalue, attained at $(0, -2, 1, 1)^T/\sqrt{6}$.

**Conclusion.** (a) $-4$; (b) $(-3, 1, 1, 1)^T/\sqrt{12}$; (c) $-10$.

**Reproducing the computer checks in 14–17.** Enter each displayed matrix using exact fractions, not rounded decimals. For a symmetric matrix, the following Python/SymPy commands return an exact spectrum and check a chosen vector. Replace `entries`, `vector`, and `eigenvalue` with the data displayed in the corresponding solution:

```
from sympy import Matrix, Rational, eye
A = Matrix(entries)
print(A.eigenvals())
v = Matrix(vector)
lam = eigenvalue
assert A == A.T
assert A*v == lam*v
u = v / (v.dot(v))**Rational(1, 2)
assert (u.T*u)[0] == 1
```

The exact eigenvector certificates above also give a complete verification without installing software. The matrices, results, and certificates are all specified; the computer problems do not require an external dataset.